

DOCUMENT #10 OF 10 — PRODUCTION CHECKLIST

Mandatory Items & System Resilience Standards

Production Verification Checklist: Loading States, Error Boundaries, Debugging, API Caching & Backend Checks

Verification Standard	Audited Stack	Compliance Level	Document Version
Zero-Regression Production Gate	React 19 + TypeScript + Supabase + LLM APIs	Production Grade (100% Mandatory)	v1.0 (Final Release)

1. Executive Verification Checklist

Mandatory Area	Engineering Standard & Implementation Pattern	Production Status
1. Loading States	Skeleton layout cards for candidate grids, upload progress indicators, streaming chat bubbles.	Enforced & Validated
2. Error Handling	React Error Boundaries on all views, toast notifications for failed API requests, fallback views.	Enforced & Validated
3. Debugging & Logging	Structured client console logger (dev-only), Supabase Edge Function error traces, Sentry capture.	Enforced & Validated
4. API Caching	LRU cache on parsed resumes, memoized LLM candidate evaluation payloads to prevent duplicate token costs.	Enforced & Validated
5. Backend Checks	PostgreSQL Row Level Security (RLS) enforcement, MIME-type pre-validation, Foreign Key cascade rules.	Enforced & Validated

2. Detailed Implementation Directives

1. Loading States & Visual Feedback

- **Skeleton Placeholders:** When querying candidate records or running vector calculations, render pulsating glassmorphic skeletons matching the 5 screen wireframes.
- **Upload Progress Tracking:** Display a dynamic 0–100% progress meter with real-time speed calculation during batch resume ingestions.
- **Optimistic State Updates:** Triage actions (Shortlist, Reject) update the UI immediately before the Supabase network response resolves.

2. Error Handling & Graceful Recovery

- **Client Error Boundaries:** Prevent entire application crashes by isolating errors to specific view cards.
- **Toast Alert System:** User-friendly notifications for upload failures, network timeouts, or invalid MIME types.
- **Zero Dead Ends:** Whenever an error occurs, provide a prominent recovery action (e.g., *"Retry Upload"* or *"Reload Candidate Grid"*).

3. Debugging & Observability Pipeline

- **Environment Guard:** Strip debug logs in production builds (`import.meta.env.PROD`).
- **Error Telemetry:** Log full stack traces from Supabase Edge Functions with correlation request IDs for rapid issue triage.

4. API Caching & Cost Optimization

- **Prompt & Entity Caching:** Hash parsed resume content so re-evaluating the same candidate against identical job criteria incurs zero LLM API cost.
- **Client State Invalidation:** React Query / Zustand invalidation only on explicit create, update, or delete triggers.

5. Backend Integrity & Security Checks

- **Pre-Storage Verification:** Validate file headers and MIME types server-side before persisting to Supabase storage buckets.
- **Multi-Tenant Isolation:** Enforce `auth.uid() = user_id` checks on all PostgreSQL table mutations to prevent cross-organization data leakage.